Submission to Global Change Biology

24 October 2026

Dear Editors,

Please consider the Research Article “Host exposure windows shape projected Septoria damage and regional sowing adaptation in European wheat” for Global Change Biology. The manuscript combines continental disease projections with explicit exposure accounting, regional sowing experiments and independent yield-evidence context. The accompanying materials preserve model definitions, observations, numerical figure data and reproduction records.

Scientific question

How do climate-driven host exposure windows organize the geography and attribution of wheat disease damage, and how do regional sowing responses depend on retained seasons and competing heat exposure?

Key findings

The 50–60°N belt supplies 68.6% of positive damage change from 44.3% of wheat area. Shared-time accounting reverses the host contribution in all three climate models. Regionally selected sowing shifts reduce the disease proxy across climate models and distinguish disease–heat tradeoffs from a uniform delay.

Importance and timeliness

Climate–disease assessments increasingly combine pathogen responses with changing crop development. Separating moving exposure windows, shared-time responses and feasible management comparisons prevents an accounting-dependent disease reduction from becoming an unsupported claim of physiological protection or grain-yield benefit.

Fit to Global Change Biology

The study links continental climate change, host development, foliar disease and agricultural exposure. Its regional responses and integration-support diagnostics address biological impact models in which environmental change alters both exposure and the period over which damage accumulates.

Biological and global-change aspects

Leaf emergence, tissue availability and symptom progression interact with three climate models and emissions pathways. Bounded sowing changes expose regional disease–heat tradeoffs. Functional canopy and yield limits remain explicit, and retrospective monitoring is distinguished from autonomous climate simulation.

Recent relevant papers

van den Bosch et al. (2026), Plant Pathology, 75, e70157, doi:10.1111/ppa.70157; Bauer et al. (2026), Global Change Biology, 32, e70776, doi:10.1111/gcb.70776; Gallego-Tévar et al. (2024), Global Change Biology, 30, e17535, doi:10.1111/gcb.17535.

Potential reviewers

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Sincerely,

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